An integer derivation cannot supply the polynomial mechanism

Polynomials
Characteristic 0
d/dX is nonzero

$$D(X^3) = 3X^2$$

Integers
Every derivation
D: $\mathbb{Z} \to \mathbb{Z}$ is zero

$$D(1) = 0$$
$$D(n) = nD(1) = 0$$

Integer ABC needs another global mechanism
Local divisibility still needs a height comparison